Visvesvaraya National Institute of Technology, Nagpur
Electronics and Communication Engineering Department
Executive M.Tech. in Applied AI
Cohort 3A sem. 3, Summer 2026
NLP Project (11th May 2026)

Financial Sentiment Analysis Agent

Presented by (Group 4)
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Presented To
Dr. Saugata Sinha,
VNIT Nagpur

Agenda

01 GitHub link, slides, and Project Report

02 Problem Statement & Dataset

03 Methodology & Pipeline

04 Models Compared

05 Training Strategy & Configuration

06 Results: PhraseBank Benchmark & Collected Data

07 Key Findings, Conclusions & Future Work

Github link, slides and Project Report

- Github link - https://github.com/milindma/ECL545_S26_NLP_Group04/ (This Repository is publicly accessible to everyone)
- Presentation link – https://github.com/milindma/ECL545_S26_NLP_Group04/tree/main/docs
- Project Report - https://github.com/milindma/ECL545_S26_NLP_Group04/tree/main/docs

Problem Statement

The Challenge

Manually analysing financial news sentiment is **time-consuming** and **inconsistent**.

The Objective

Build an automated system to classify headlines into **Positive**, **Negative**, or **Neutral** sentiment.

Approaches Evaluated:

- BERT (zero-shot)
- Fine-tuned BERT
- FinBERT (zero-shot)

Dataset Overview

Training Data

Financial headlines (financial_sentiment.csv) merged from multiple public sources.

Split: 80/10/10 (train/val/test)

Evaluation Set 1

453 sentences with the following breakdown

Breakdown: 114(+), 61(-), 278(n)

Evaluation Set 2 — Collected Unseen Data

1,845 financial headlines from Economic Times & The Hindu Business Line (April 2026).

Balanced evaluation subset of 1,000 samples (334/333/333) for fair per-class assessment.

Labels: Positive, Negative, Neutral *(Manually verified)*

Methodology & Pipeline

A four-stage pipeline was implemented to systematically compare zero-shot and fine-tuned transformer models for financial sentiment classification.

01

Data Preparation

→

02

Baseline Evaluation (BERT & FinBERT zero-shot)

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03

Fine-Tuning BERT (weighted loss, cosine decay, early stopping)

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04

Three-Way Evaluation (Training Dataset + Collected Data)

→

05

Analysis & Reporting

Models & Training Configuration

Models Compared

- **BERT (Zero-shot):** bert-base-uncased— no training; lower-bound baseline
- **BERT (Fine-tuned):** bert-base-uncased + classification head; fine-tuned on financial_sentiment.csv; class-weighted cross-entropy loss
- **FinBERT (Zero-shot):** ProsusAI/finbert; finance-domain pre-trained; zero-shot

Key Training Hyperparameters

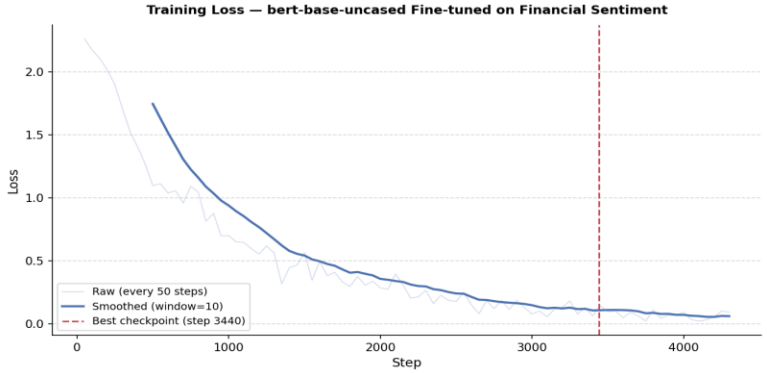
Optimization: Learning Rate: 1e-5 | Epochs: 10 max (early stopping) | Batch Size: 8 (effective 16)

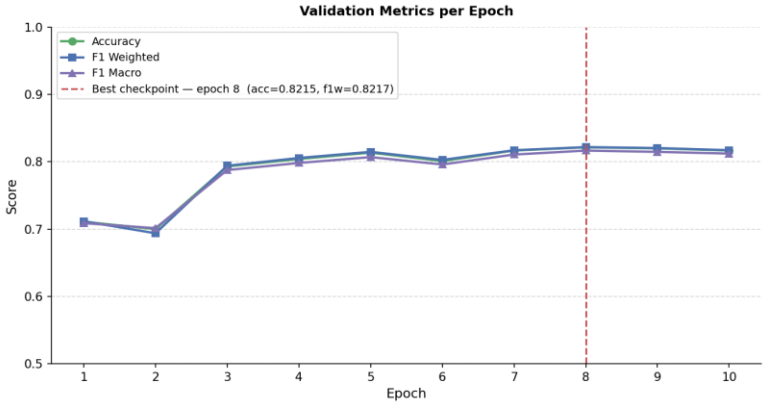
Architecture: LR Scheduler: Cosine Decay (10% warmup) | Weight Decay: 0.01 | Max Seq: 128

Best Checkpoint: Epoch 8 (Step 3440) — Val Acc: 82.15%, Val F1 Weighted: 82.17%

Training Dynamics & Insights

Best Checkpoint: Epoch 8 (Step 3440) — **Val Accuracy: 82.15%, Val F1 Weighted: 82.17%**





Strategy & Improvements

- Early stopping (patience=3) monitored Val F1 Weighted; prevented overfitting

Results and Demo

- FinBERT (97.6% acc) marginally leads Fine-tuned BERT (95.8%) — both far exceed BERT zero-shot (27.4%)
- Fine-tuned BERT: strong across all classes (Pos F1=0.92, Neg F1=0.95, Neu F1=0.97)

During First Evaluation

Metric	Fine-tuned v1	Fine-tuned v2
Accuracy	0.33	0.38
Weighted F1	0.32	0.37

Today:

Model	Accuracy	F1 Macro	F1 Weighted
BERT (Zero-shot)	0.274	0.237	0.144
BERT (Fine-tuned)	0.958	0.949	0.958
FinBERT (Zero-shot)	0.976	0.965	0.976

Collected Data

Collected_Test_Data.csv

Open with Numbers

Source	Date	Text
economictimes.indiatimes.com	2026-04-24 12:54:56	Sebi clears 4 IPOs including Yatayat Corporation and EAAA India Alternatives. Sebi has approved IPOs for four firms spanning logistics, alternatives, real estate and manufacturing. Companies aim to tap rising
livemint.com	2026-04-24 12:49:15	Small-cap stock under ₹50 Aayush Wellness jumps 9% despite bloodbath on Dalal Street. Aayush Wellness shares rose 9% to ₹39, hitting a three-month high, with a cumulative gain of 21% over two days. Th
thehindubusinessline.com	2026-04-24 12:40:15	IT rout and crude surge drag Nifty to third straight loss. Nifty 50 sheds 275 points or 1.14% to end at 23,898, Sensex tumbles 983 points to close at 76,681
economictimes.indiatimes.com	2026-04-24 12:37:21	Buy the dip, but avoid IT, back pharma & power, says Rudramurthy BV. Indian markets are experiencing a healthy correction, according to Vachana Investments' MD, CA Rudramurthy BV. He advises waitir
economictimes.indiatimes.com	2026-04-24 12:32:00	Quote of the day by John Maynard Keynes: "The actual results of an investment over a long term of years very seldom agree with the initial expectation.". John Maynard Keynes' insight that investment results
economictimes.indiatimes.com	2026-04-24 12:23:57	When billionaire Nithin Kamath wanted to earn just Rs 5 crore and retire in Goa. Nithin Kamath warns wealth inequality, driven by asset inflation and concentrated ownership, risks social strain. Reflecting on his
economictimes.indiatimes.com	2026-04-24 12:12:04	US Stock Market Today | Dow Jones | Nasdaq Live: US futures mixed as Iran war deadlock keeps investors edgy.
economictimes.indiatimes.com	2026-04-24 12:10:47	Sebi eases FPI settlement norms, allows netting of funds in cash market to cut costs, improve efficiency. Sebi has allowed net settlement of funds for foreign portfolio investors in the cash market to reduce cos
thehindubusinessline.com	2026-04-24 12:02:43	Q4 Results Today Live: Adani Green Energy Q4 con. profit soars, Shriram Finance, Hindustan Zinc, IndusInd & Chennai Petroleum Q4 profit rise, Reliance, M&M Financial, Lodha Developers to announce Q4 resu
thehindubusinessline.com	2026-04-24 11:40:09	Global bonds set for worst week in a month as US-Iran risks rise. Investors have questioned in recent weeks whether central banks would deliver on the aggressive tightening priced by markets, but renewed u
thehindubusinessline.com	2026-04-24 11:40:05	JSW Steel, Japan's JFE formalise 50:50 joint venture 'JSW JFE Steel Ltd' in Odisha. JSW Steel closes Friday at ₹1,255.70, down 0.10% on the day
economictimes.indiatimes.com	2026-04-24 11:39:11	Polymarket gambler makes $35,000 profit by using hair dryer to tamper with temperature readings at Paris airport. A Polymarket user allegedly made $35,000 by exploiting unusual temperature spikes recorde
thehindubusinessline.com	2026-04-24 11:30:55	Infosys shares fall 7% after Q4, muted guidance weighs. Infosys guided for FY27 revenue growth of 1.5 per cent to 3.5 per cent, indicating a subdued demand environment
thehindubusinessline.com	2026-04-24 11:30:04	Rupee ends 22 paise weaker at 94.23 against US dollar. High crude oil prices and strengthening US dollar pulled the rupee down

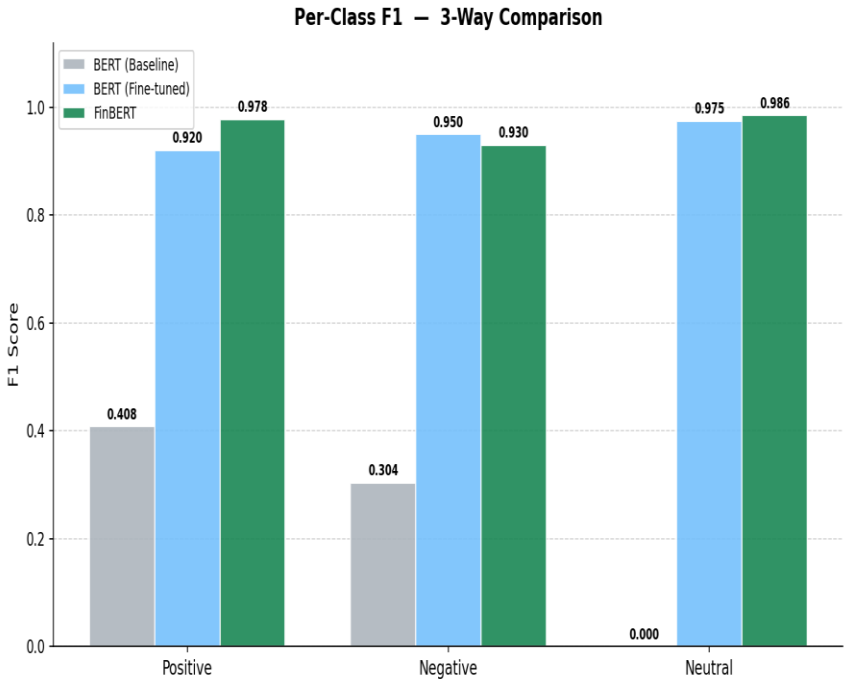
Results — Collected Unseen Data (1,000 balanced samples)

- Real-world test: 334 positive / 333 negative / 333 neutral — balanced for fair evaluation
- Fine-tuned BERT leads: 72.9% accuracy, 71.7% F1 Weighted — outperforms FinBERT by ~13pp
- FinBERT struggles on Neutral (F1=0.379) despite domain pre-training; fine-tuned BERT achieves 0.591

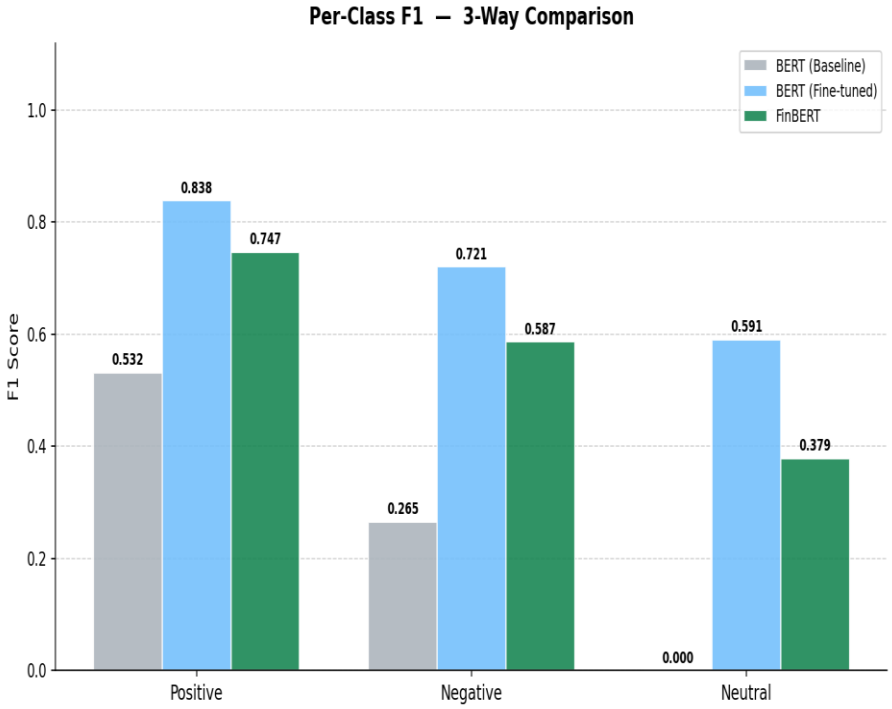
Class	BERT Zero-shot	BERT Fine-tuned	FinBERT
Negative	0.265	0.721	0.587
Neutral	0.000	0.591	0.379
Positive	0.532	0.838	0.747

Model	BERT (Fine-tuned)	F1 Macro	F1 Weighted
Accuracy	0.729	0.716	0.717
FinBERT (Zero-shot)	0.594	0.571	0.571

Comparative Analysis

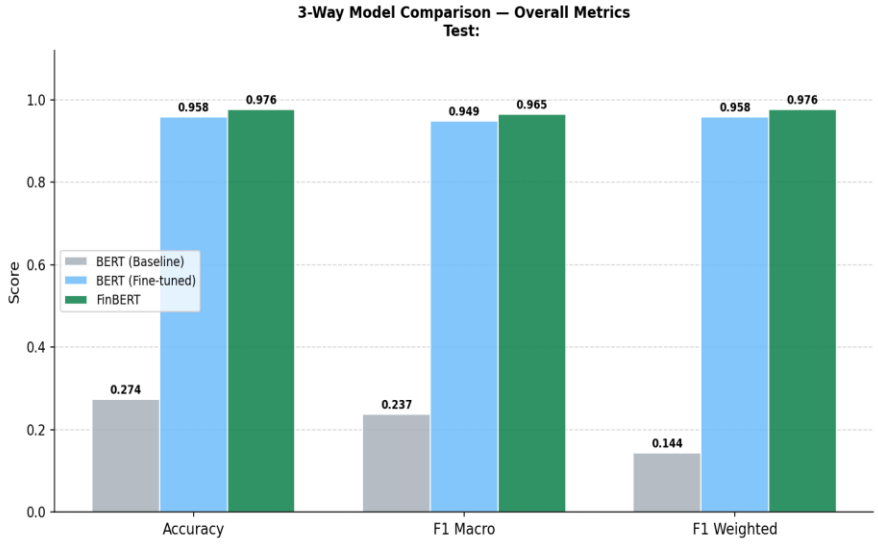


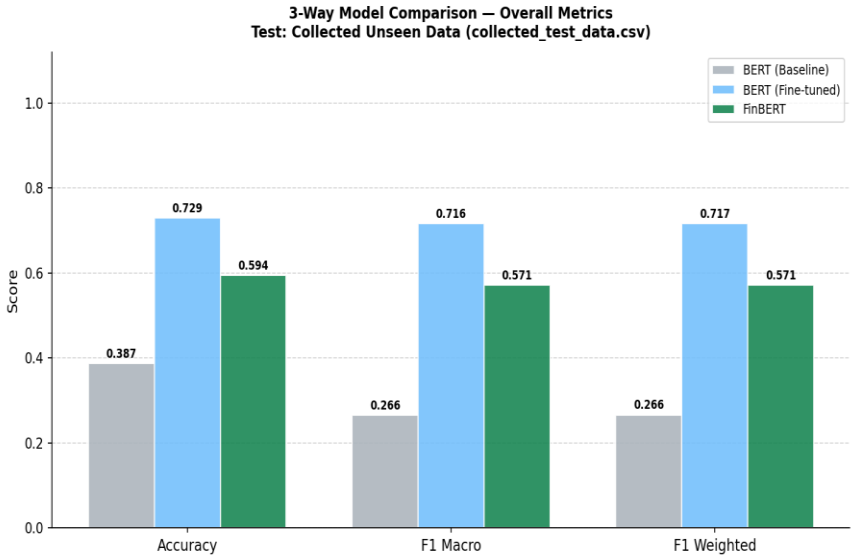
Test Data



Collected Data

Comparative Analysis

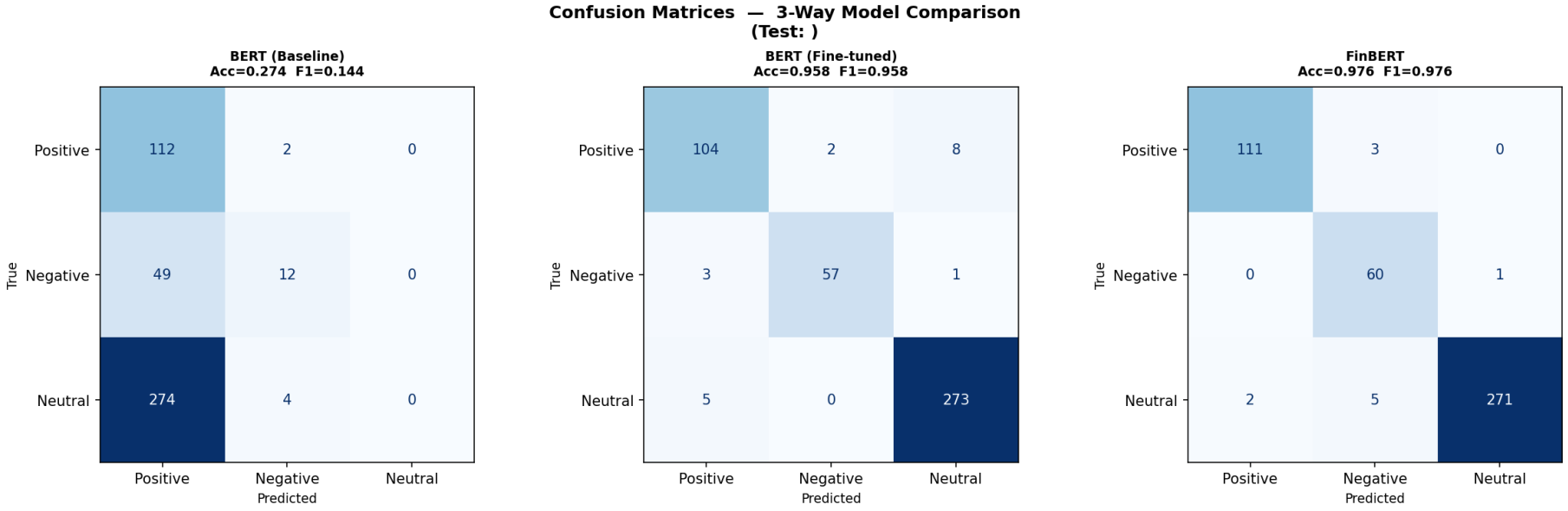




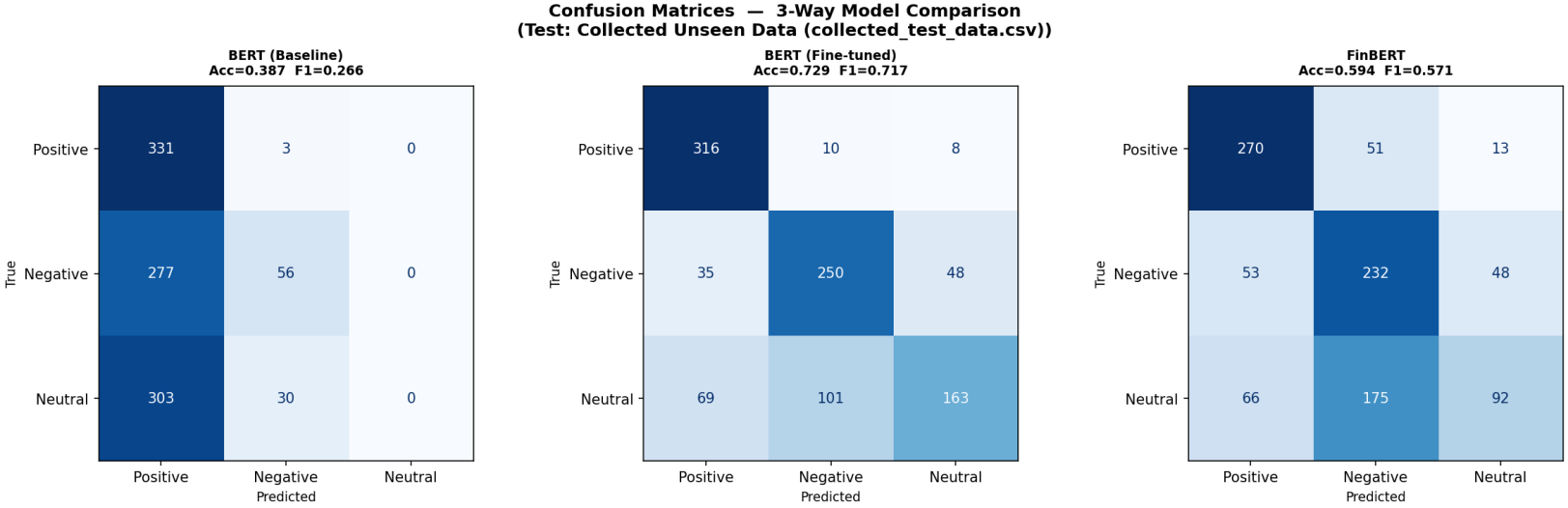
Test Data

Collected Data

Confusion Matrices (Test Data)



Confusion Matrices (Collected Data)



THANK YOU